# VNR VJIET

**Name of the Laboratory:**

**Name of the Experiment:**_____

**Experiment No:**_____ **Date:**_____

## WEEK-1

**1. Program to implement FCFS scheduling algorithm without arrival times.**

**Aim:** To implement FCFS scheduling algorithm without Arrival times given and finding average Waiting time and average Turnaround time.

## Algorithm:

1. **Start.**
2. Read the number of processes n.
3. Enter the **Burst Time (BT)** for each process.
4. Set the **Waiting Time (WT)** of the first process to 0.
5. For each remaining process:

   WT[i] = WT[i-1] + BT[i-1]
6. Calculate the **Turnaround Time (TAT)** for each process:

   TAT[i] = WT[i] + BT[i]
7. Calculate:
   - **Average Waiting Time (AWT)** = Sum of Waiting Time / n
   - **Average Turnaround Time (ATAT)** = Sum of Turnaround Time / n
8. Display the **Process ID, Burst Time, Waiting Time, and Turnaround Time**.
9. **Stop.**

**Problem:**

Consider the following process and calculate average WT and average TAT. Assume that the processes arrived in the order as follows:

| Process | - | Burst Time |
|---|---|---|
| P1 | - | 5 |
| P2 | - | 3 |
| P3 | - | 2 |

# VNR VJIET

**Name of the Laboratory:**

**Name of the Experiment:**_____

**Experiment No:**_____ **Date:**_____

**Program:**

```
#include <stdio.h>

int main() {
   int n, i;

   printf("Enter number of processes: ");
   scanf("%d", &n);

   int bt[n], wt[n], tat[n];
   float avg_wt = 0, avg_tat = 0;


   printf("Enter Burst Time for each process:\n");
   for(i = 0; i < n; i++) {
      printf("P%d: ", i + 1);
      scanf("%d", &bt[i]);
   }


   wt[0] = 0;
   for(i = 1; i < n; i++) {
      wt[i] = wt[i - 1] + bt[i - 1];
   }


   for(i = 0; i < n; i++) {
      tat[i] = wt[i] + bt[i];

      avg_wt += wt[i];
      avg_tat += tat[i];
   }


   printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
   for(i = 0; i < n; i++) {
      printf("P%d\t%d\t\t%d\t\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
   }

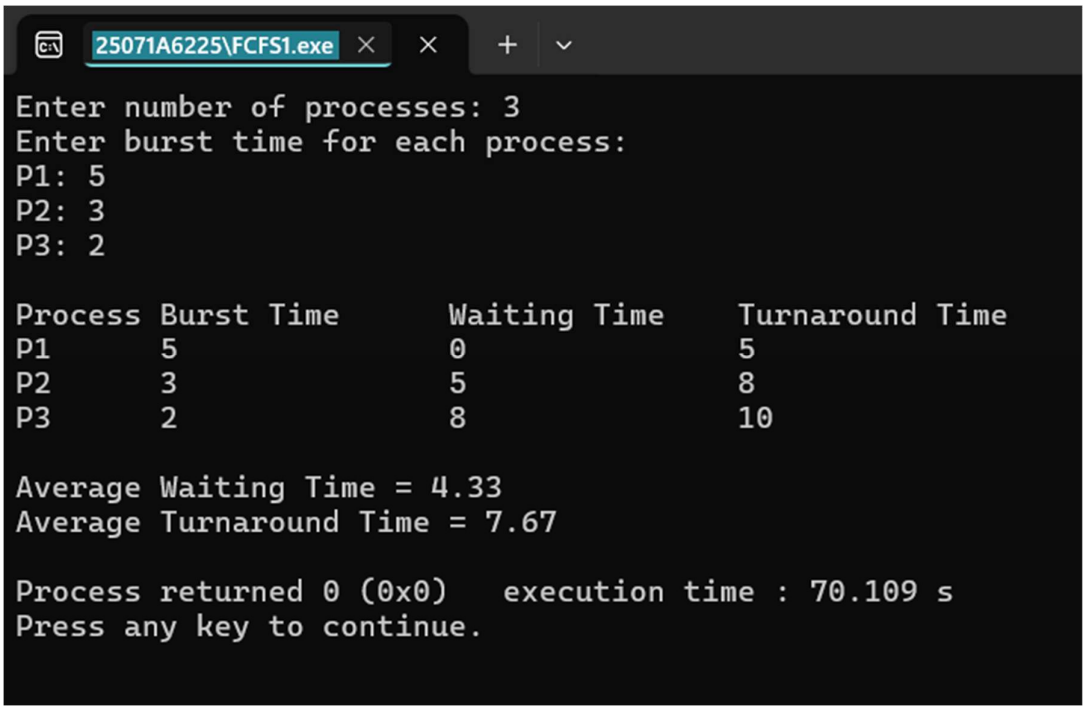
   avg_wt /= n;
   avg_tat /= n;

   printf("\nAverage Waiting Time = %.2f", avg_wt);
   printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
```

```
    return 0;
}
```

**OUTPUT:**

**TESTCASE 1:**



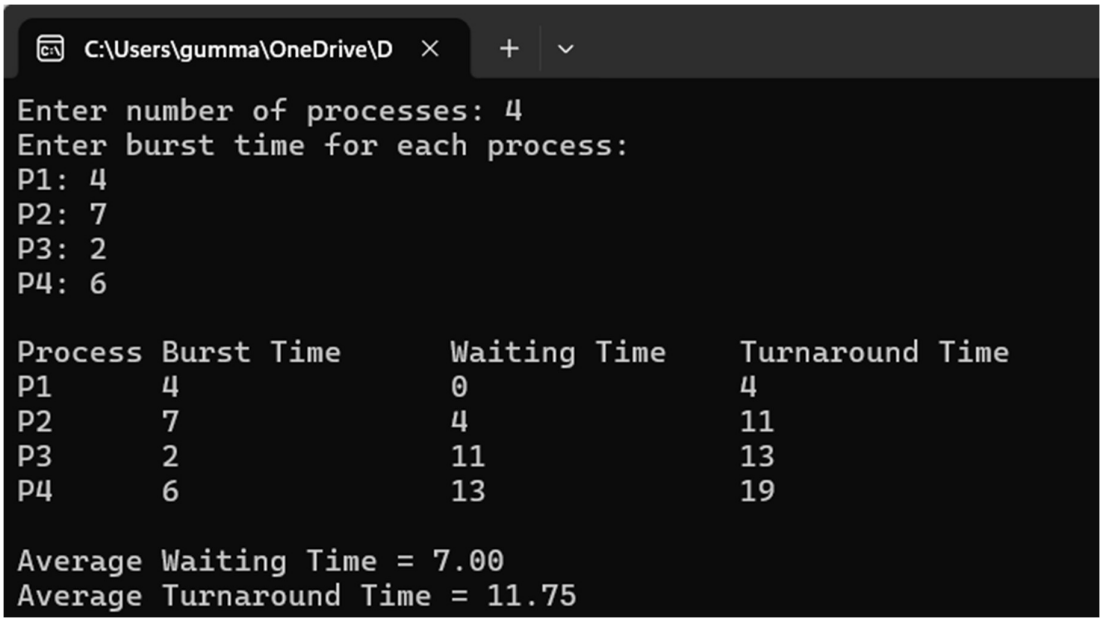
```
25071A6225\FCFS1.exe
Enter number of processes: 3
Enter burst time for each process:
P1: 5
P2: 3
P3: 2

Process Burst Time      Waiting Time    Turnaround Time
P1      5               0               5
P2      3               5               8
P3      2               8               10

Average Waiting Time = 4.33
Average Turnaround Time = 7.67

Process returned 0 (0x0)   execution time : 70.109 s
Press any key to continue.
```

**TESTCASE 2:**



```
C:\Users\gumma\OneDrive\D
Enter number of processes: 4
Enter burst time for each process:
P1: 4
P2: 7
P3: 2
P4: 6

Process Burst Time      Waiting Time    Turnaround Time
P1      4               0               4
P2      7               4               11
P3      2               11              13
P4      6               13              19

Average Waiting Time = 7.00
Average Turnaround Time = 11.75
```

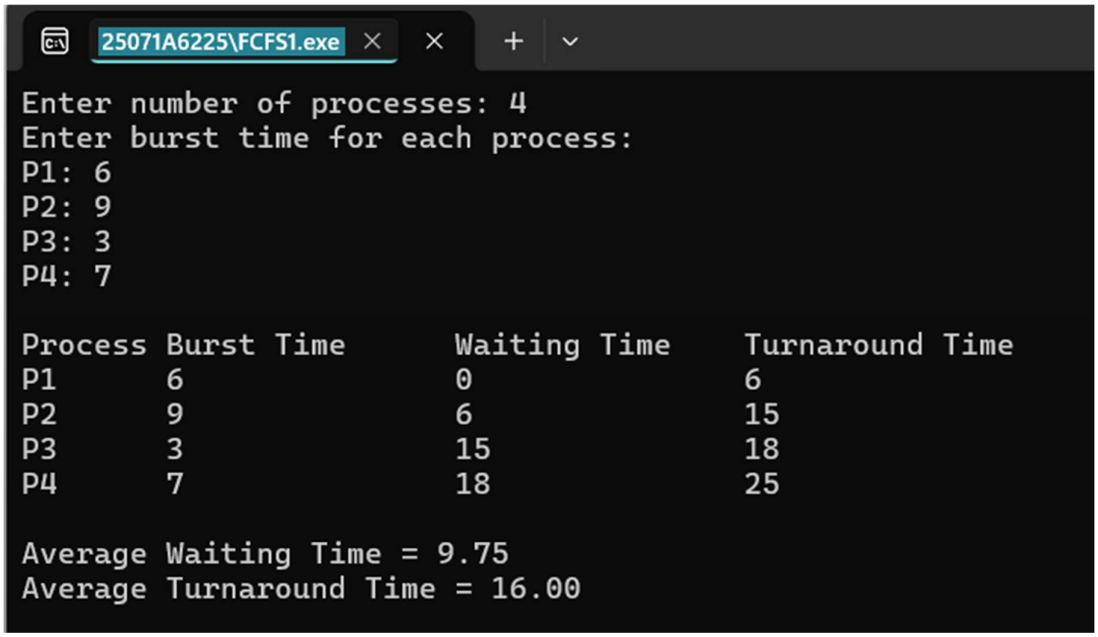
# VNR VJIET

**Name of the Laboratory:**

**Name of the Experiment:**_____

**Experiment No:**_____ **Date:**_____

**TESTCASE 3:**



**2. Program to implement FCFS scheduling algorithm with arrival times.**

**Aim:** To implement FCFS scheduling algorithm with Arrival times given and finding average Waiting time and average Turnaround time.

**Algorithm:**

1. Start.

2.Enter the number of processes n.

Enter the **Arrival Time (AT)** and **Burst Time (BT)** for each process.

Arrange the processes in ascending order of their arrival time.

Set the current time to 0.

For each process:

- If the current time is less than its arrival time, set current time equal to its arrival time.
- Calculate **Completion Time (CT)**:
  CT = Current Time + Burst Time
- Calculate **Turnaround Time (TAT)**:
  TAT = CT - Arrival Time
- Calculate **Waiting Time (WT)**:
  WT = TAT - Burst Time
- Update current time to the completion time.

Calculate average waiting time and average turnaround time.

Display the process details and averages.

Stop.

# VNR VJIET

**Name of the Laboratory:**

**Name of the Experiment:**_____

**Experiment No:**_____ **Date:**_____

**Problem:** Calculate the average waiting time and average turnaround time for processes given below

| Process | AT | BT |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 2 | 8 |
| P3 | 3 | 9 |
| P4 | 5 | 7 |

**Program:**

```
#include <stdio.h>

int main() {
   int n, i, j;
   int at[20], bt[20], ct[20], tat[20], wt[20];
   float avgWT = 0, avgTAT = 0;

   printf("Enter number of processes: ");
   scanf("%d", &n);

   printf("\nEnter Arrival Time and Burst Time:\n");
   for(i = 0; i < n; i++) {
      printf("Process P%d\n", i + 1);
      printf("Arrival Time: ");
      scanf("%d", &at[i]);
      printf("Burst Time: ");
      scanf("%d", &bt[i]);
   }

   for(i = 0; i < n - 1; i++) {
      for(j = i + 1; j < n; j++) {
         if(at[i] > at[j]) {
            int temp;
            temp = at[i];
            at[i] = at[j];
            at[j] = temp;

            temp = bt[i];
            bt[i] = bt[j];
            bt[j] = temp;
         }
      }
   }

   ct[0] = at[0] + bt[0];


 for(i = 1; i < n; i++) {
      if(ct[i - 1] < at[i])
```

```
        ct[i] = at[i] + bt[i];




        else
                ct[i] = ct[i - 1] + bt[i];
           }


           for(i = 0; i < n; i++) {
              tat[i] = ct[i] - at[i];
              wt[i] = tat[i] - bt[i];

              avgWT += wt[i];
              avgTAT += tat[i];
           }

           avgWT /= n;
           avgTAT /= n;

           printf("\nP\tAT\tBT\tCT\tTAT\tWT\n");
           for(i = 0; i < n; i++) {
              printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
                    i + 1, at[i], bt[i], ct[i], tat[i], wt[i]);
           }

           printf("\nAverage Waiting Time = %.2f", avgWT);
           printf("\nAverage Turnaround Time = %.2f\n", avgTAT);

           return 0;
}
```

**OUTPUT:**

**TESTCASE 1:**

```
25071A6225\FCFS2.exe
Enter Burst Time for P3: 9

Enter Arrival Time for P4: 5
Enter Burst Time for P4: 7

------------------------------------------------
Process AT      BT      CT      WT      TAT
------------------------------------------------
P1      0       5       5       0       5
P2      2       8       13      3       11
P3      3       9       22      10      19
P4      5       7       29      17      24
------------------------------------------------
Average Waiting Time = 7.50
Average Turnaround Time = 14.75
```

**TESTCASE 2:**

```
C:\Users\gumma\OneDrive\D
Enter number of processes: 3

Enter Arrival Time for P1: 5
Enter Burst Time for P1: 6

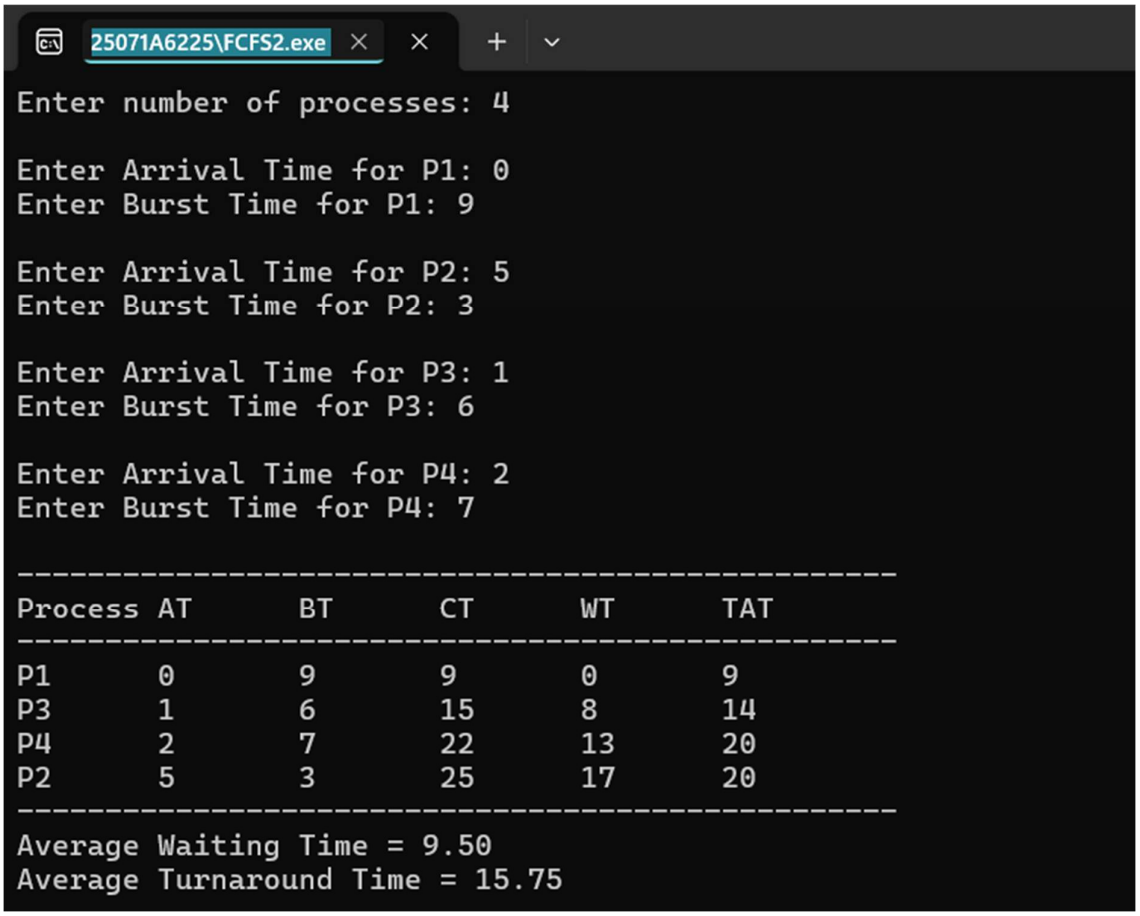
Enter Arrival Time for P2: 0
Enter Burst Time for P2: 4

Enter Arrival Time for P3: 2
Enter Burst Time for P3: 9

------------------------------------------------
Process AT      BT      CT      WT      TAT
------------------------------------------------
P2      0       4       4       0       4
P3      2       9       13      2       11
P1      5       6       19      8       14
------------------------------------------------
Average Waiting Time = 3.33
Average Turnaround Time = 9.67
```

H

**TESTCASE 3:**


25071A6225\FCFS2.exe
Enter number of processes: 4
Enter Arrival Time for P1: 0
Enter Burst Time for P1: 9
Enter Arrival Time for P2: 5
Enter Burst Time for P2: 3
Enter Arrival Time for P3: 1
Enter Burst Time for P3: 6
Enter Arrival Time for P4: 2
Enter Burst Time for P4: 7
Process AT BT CT WT TAT
P1 0 9 9 0 9
P3 1 6 15 8 14
P4 2 7 22 13 20
P2 5 3 25 17 20
Average Waiting Time = 9.50
Average Turnaround Time = 15.75

**3. Program to implement SJF scheduling algorithm in preemptive mode (SRTF).**

**Aim:** To implement SJF scheduling algorithm in preemptive mode given and finding average Waiting time and average Turnaround time.

## Algorithm:

1. Start.
2. Read the number of processes n.
3. Read the Arrival Time (AT) and Burst Time (BT) for each process.
4. Set the current time t = 0.

5 At each unit of time:

- Find all processes that have arrived (AT ≤ t) and are not completed.
- Select the process with the smallest remaining burst time.
- If a new process arrives with a shorter remaining time, preempt the currently running process and execute the new one.

6. Reduce the selected process's remaining burst time by 1.
7. If the remaining burst time becomes 0:

Mark the process as completed.

Calculate:

- Completion Time (CT) = Current Time + 1
- Turnaround Time (TAT) = CT − AT
- Waiting Time (WT) = TAT − BT

 Repeat Steps 5–7 until all processes are completed.

 Calculate:

- Average Waiting Time = ΣWT / n
- Average Turnaround Time = ΣTAT / n

 Stop.

**Problem:** Calculate the average waiting time and average turnaround time for processes given below

| Process | AT | BT |
|---|---|---|
| P1 | 0 | 3 |
| P2 | 1 | 6 |
| P3 | 2 | 4 |
| P4 | 3 | 2 |
| P5 | 4 | 7 |

# VNR VJIET

**Name of the Laboratory:**

**Name of the Experiment:**_____

**Experiment No:**_____ **Date:**_____

**Program :**

```
#include <stdio.h>
int main() {
    int n, i, time = 0, completed = 0;
    int at[20], bt[20], rt[20];
    int ct[20], tat[20], wt[20];
    int shortest, min;
    float avgWT = 0, avgTAT = 0;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++) {
        printf("Enter Arrival Time and Burst Time of P%d: ", i + 1);
        scanf("%d%d", &at[i], &bt[i]);
        rt[i] = bt[i];
    }
    while(completed < n) {
       shortest = -1;
        min = 9999;

        for(i = 0; i < n; i++) {
            if(at[i] <= time && rt[i] > 0 && rt[i] < min) {
                min = rt[i];
                shortest = i;
            }
        }

        if(shortest == -1) {
            time++;
            continue;
        }
        rt[shortest]--;
        time++;
        if(rt[shortest] == 0) {
            completed++;
            ct[shortest] = time;
            tat[shortest] = ct[shortest] - at[shortest];
            wt[shortest] = tat[shortest] - bt[shortest];
```

```
            avgWT += wt[shortest];

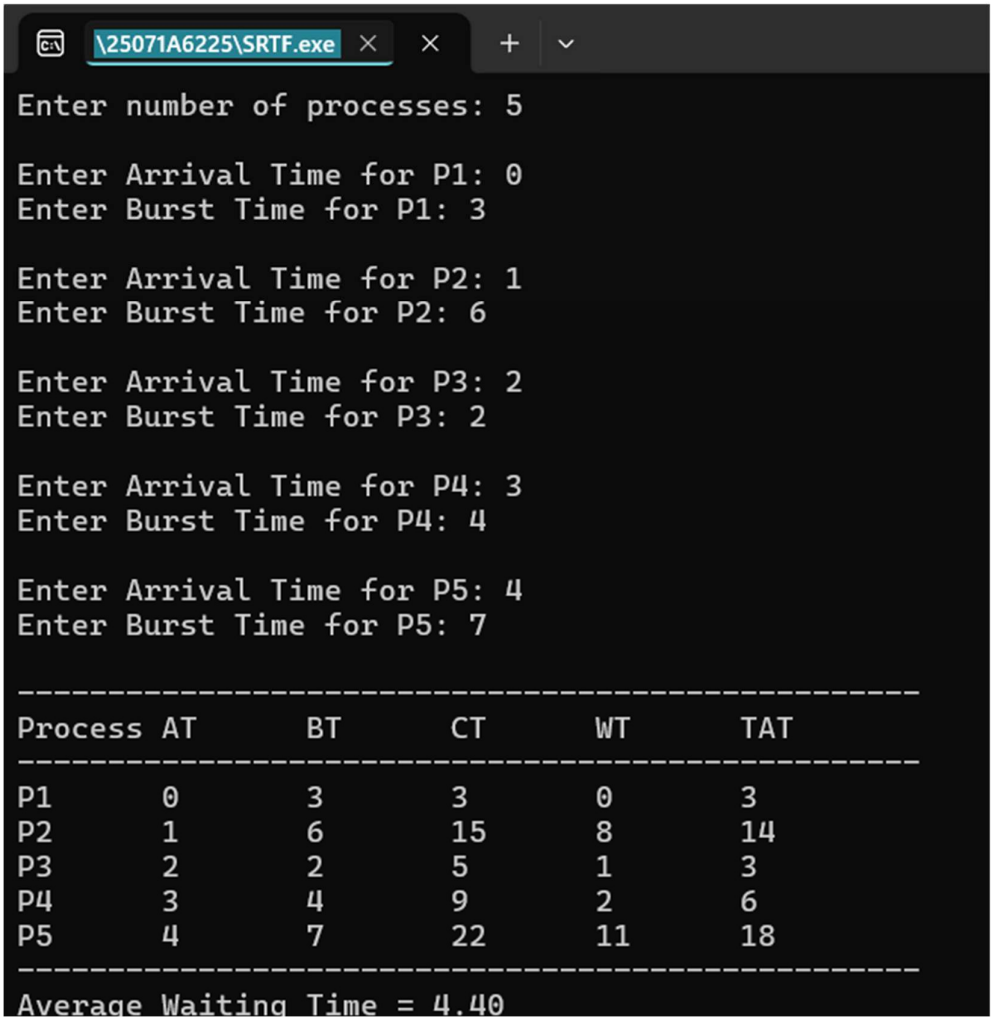


    avgTAT += tat[shortest];
          }
       }
    printf("\n----------------------------------------------------\n");
       printf("Process\tAT\tBT\tCT\tTAT\tWT\n");
       printf("----------------------------------------------------\n");

       for(i = 0; i < n; i++) {
          printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
                i + 1, at[i], bt[i], ct[i], tat[i], wt[i]);
       }
       printf("----------------------------------------------------\n");
       printf("Average Waiting Time = %.2f\n", avgWT / n);
       printf("Average Turnaround Time = %.2f\n", avgTAT / n);
       return 0;
}
```

**OUTPUT:**

**TEST CASE 1:**


\25071A6225\SRTF.exe
Enter number of processes: 5
Enter Arrival Time for P1: 0
Enter Burst Time for P1: 3
Enter Arrival Time for P2: 1
Enter Burst Time for P2: 6
Enter Arrival Time for P3: 2
Enter Burst Time for P3: 2
Enter Arrival Time for P4: 3
Enter Burst Time for P4: 4
Enter Arrival Time for P5: 4
Enter Burst Time for P5: 7
Process AT BT CT WT TAT
P1 0 3 3 0 3
P2 1 6 15 8 14
P3 2 2 5 1 3
P4 3 4 9 2 6
P5 4 7 22 11 18
Average Waiting Time = 4.40

**TESTCASE 2:**

```
\25071A6225\SRTF.exe
Enter number of processes: 3

Enter Arrival Time for P1: 2
Enter Burst Time for P1: 6

Enter Arrival Time for P2: 0
Enter Burst Time for P2: 9

Enter Arrival Time for P3: 1
Enter Burst Time for P3: 7

------------------------------------------------
Process AT      BT      CT      WT      TAT
------------------------------------------------
P1      2       6       8       0       6
P2      0       9       22      13      22
P3      1       7       14      6       13
------------------------------------------------
Average Waiting Time = 6.33
Average Turnaround Time = 13.67
```

**TESTCASE 3:**

```
\25071A6201\sjfpre.exe
Enter number of processes: 5
Enter Arrival Time and Burst Time of P1: 0 7
Enter Arrival Time and Burst Time of P2: 1 5
Enter Arrival Time and Burst Time of P3: 2 3
Enter Arrival Time and Burst Time of P4: 3 1
Enter Arrival Time and Burst Time of P5: 4 2

------------------------------------------------
Process AT      BT      CT      TAT     WT
------------------------------------------------
P1      0       7       18      18      11
P2      1       5       12      11      6
P3      2       3       6       4       1
P4      3       1       4       1       0
P5      4       2       8       4       2
------------------------------------------------
Average Waiting Time = 4.00
Average Turnaround Time = 7.60

Process returned 0 (0x0)   execution time : 29.241 s
Press any key to continue.
```

**4. Program to implement SJF scheduling algorithm in non-preemptive mode.**

**Aim:** To implement SJF scheduling algorithm in non-preemptive mode given and finding average Waiting time and average Turnaround time.

## Algorithm:

1. Start.
2. Read the number of processes n.
3. Read the Arrival Time (AT) and Burst Time (BT) for each process.
4. Set the current time t = 0 and completed processes count = 0.
5. While all processes are not completed:
   - Find all processes that have arrived (AT ≤ t) and are not completed.
   - Select the process with the smallest Burst Time (BT).
   - If no process has arrived, increment the current time by 1.

Otherwise:

- Execute the selected process until it finishes (without interruption).
- Update:
  - Completion Time (CT) = t + BT
  - Turnaround Time (TAT) = CT − AT
  - Waiting Time (WT) = TAT − BT
- Mark the process as completed.
- Update the current time to the completion time.

6. Repeat Step 5 until all processes are completed.
7. Calculate:
   - Average Waiting Time = ΣWT / n

- Average Turnaround Time = ΣTAT / n

9. Display Process ID, AT, BT, CT, TAT, and WT.

10. Stop.

## Problem:

Calculate Average waiting time , Average turn around time using SRTF for the following processes given below.

| Process | AT | BT |
|---|---|---|
| P1 | 0 | 8 |
| P2 | 1 | 4 |
| P3 | 2 | 7 |
| P4 | 3 | 1 |

# VNR VJIET

**Name of the Laboratory:**

**Name of the Experiment:**_____

**Experiment No:**_____ **Date:**_____

**Program:**

```
#include <stdio.h>


    int main() {
        int n, i, time = 0, completed = 0;
        int at[20], bt [20], ct[20], tat[20], wt[20];
        int visited[20] = {0};
       printf("Enter number of processes: ");
        scanf("%d", &n);


        for(i = 0; i < n; i++) {
            printf("Enter Arrival Time of P%d: ", i + 1);
            scanf("%d", &at[i]);


            printf("Enter Burst Time of P%d: ", i + 1);
            scanf("%d", &bt[i]);
        }
    while(completed < n) {
            int idx = -1;
            int minBT = 9999;


            for(i = 0; i < n; i++) {
                if(at[i] <= time && visited[i] == 0 && bt[i] < minBT) {
                    minBT = bt[i];
                    idx = i;
                }
            }
```

```
        if(idx == -1) {
            time++;


    } else {
            ct[idx] = time + bt[idx];
            time = ct[idx];
            tat[idx] = ct[idx] - at[idx];
            wt[idx] = tat[idx] - bt[idx];
            visited[idx] = 1;
            completed++;
        }
      }

      float avgWT = 0, avgTAT = 0;

      printf("Process\tAT\tBT\tCT\tWT\tTAT\n");

      for(i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
              i + 1, at[i], bt[i], ct[i], wt[i], tat[i]);
        avgWT += wt[i];
        avgTAT += tat[i];
      }

      printf("Average Waiting Time = %.2f", avgWT / n);
      printf("Average Turnaround Time = %.2f\n", avgTAT / n);

      return 0;}
```

# VNR VJIET

**Name of the Laboratory:**

**Name of the Experiment:**_____

**Experiment No:**_____ **Date:**_____

**OUTPUT:**

**TEST CASE 1:**

```
\25071A6225\SRTF.exe
Enter number of processes: 4
Enter Arrival Time for P1: 0
Enter Burst Time for P1: 8
Enter Arrival Time for P2: 1
Enter Burst Time for P2: 4
Enter Arrival Time for P3: 2
Enter Burst Time for P3: 7
Enter Arrival Time for P4: 3
Enter Burst Time for P4: 1

------------------------------------------------
Process AT      BT      CT      WT      TAT
------------------------------------------------
P1      0       8       13      5       13
P2      1       4       6       1       5
P3      2       7       20      11      18
P4      3       1       4       0       1
------------------------------------------------
Average Waiting Time = 4.25
Average Turnaround Time = 9.25
```

**TESTCASE 2:**

```
nts\25071A6225\sjf.exe
Enter number of processes: 3
Enter Arrival Time of P1: 0
Enter Burst Time of P1: 5
Enter Arrival Time of P2: 1
Enter Burst Time of P2: 3
Enter Arrival Time of P3: 2
Enter Burst Time of P3: 1

Process AT      BT      CT      WT      TAT
P1      0       5       5       0       5
P2      1       3       9       5       8
P3      2       1       6       3       4

Average Waiting Time = 2.67
Average Turnaround Time = 5.67
```

**TESTCASE 3:**

```
nts\25071A6225\sjf.exe
Enter Burst Time of P1: 3
Enter Arrival Time of P2: 4
Enter Burst Time of P2: 2
Enter Arrival Time of P3: 5
Enter Burst Time of P3: 1
Enter Arrival Time of P4: 6
Enter Burst Time of P4: 4

Process AT      BT      CT      WT      TAT
P1      2       3       5       0       3
P2      4       2       8       2       4
P3      5       1       6       0       1
P4      6       4       12      2       6

Average Waiting Time = 1.00
Average Turnaround Time = 3.50

Process returned 0 (0x0)   execution time : 27.520 s
Press any key to continue.
```